

DATA 308: Lab 3a

Sampling Strategies

Simulate sampling

Here we will practice the various sampling strategies to collect samples of size $n = 20$.

Proceed to do each of the sampling strategies we learned. Circle a person to indicate that they are included in the sample.

- For convenience sampling, choose one place to “stand”. The nearest 20 people are your sample.
- For random sampling, use R to randomly sample numbers, which correspond to the position of the people that will be in your sample. Set your seed to be your birthday (month and day).
 - For cluster and multistage sampling, choose 2 groups to sample.

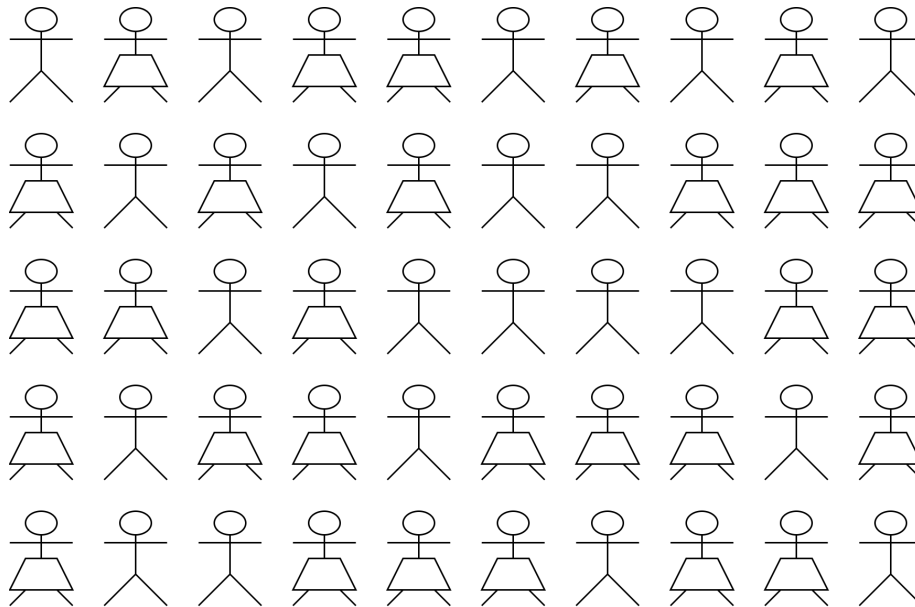


Figure 1: convenience sample

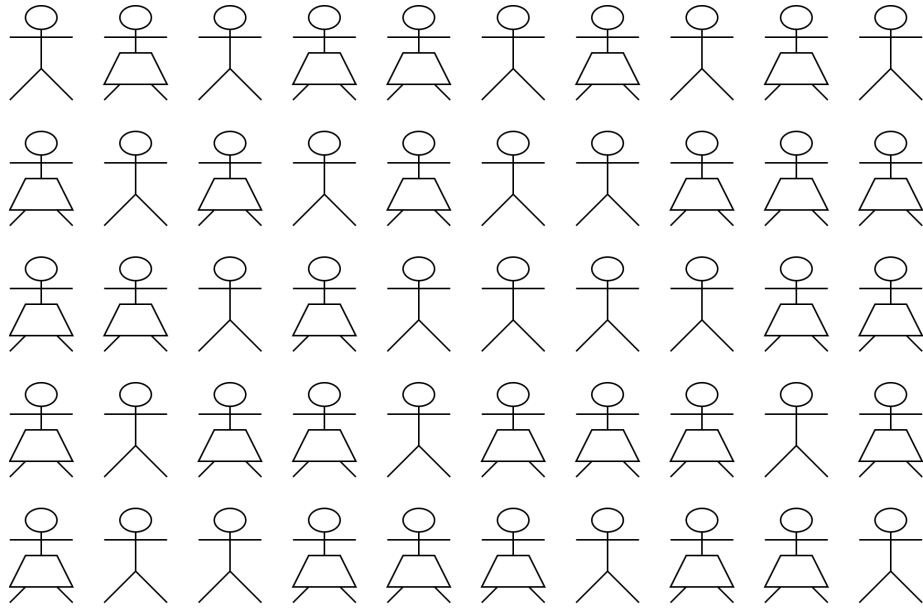


Figure 2: Simple random sample

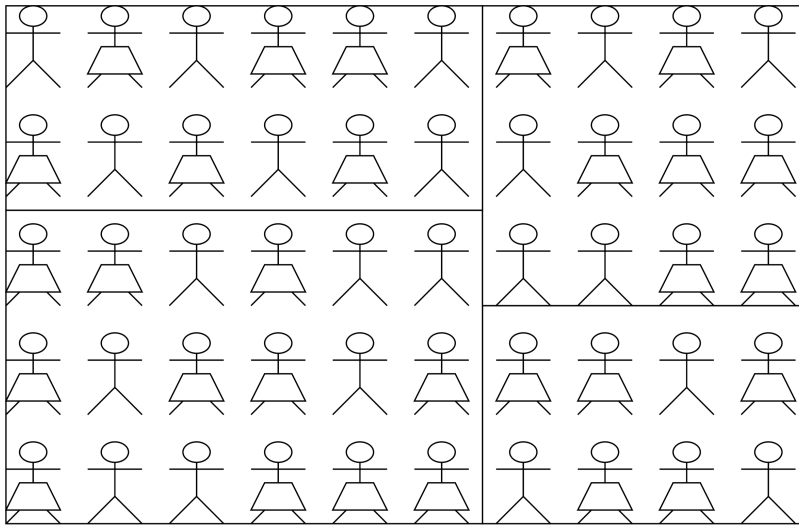


Figure 3: Stratified sample

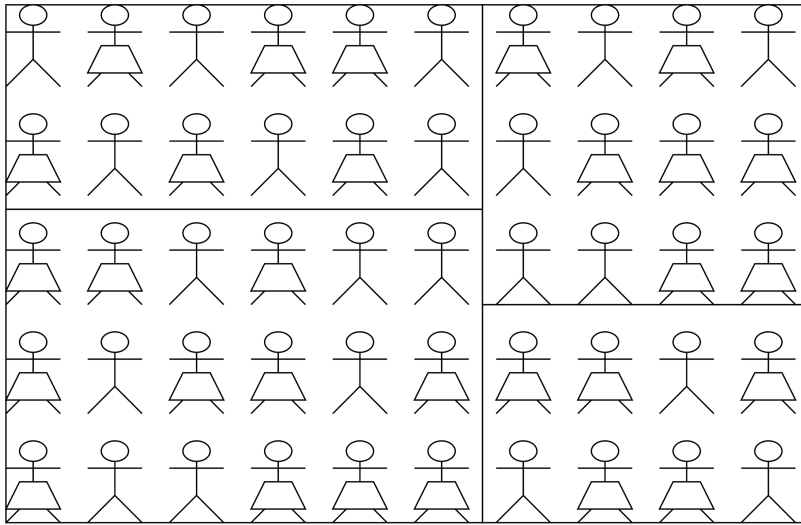


Figure 4: Cluster sampling

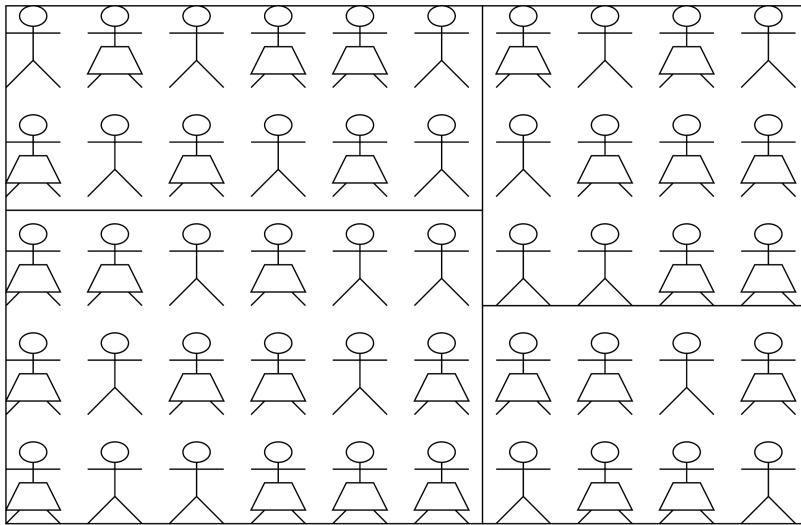


Figure 5: Multistage sampling

Reflection questions

Question 1

What was your population proportion of people wearing a trapezoid?

Question 2

What was your sample proportion of people wearing a trapezoid, for your simple random sample?

Question 3

Ask 5 people in class what their sample proportions were and report them here.

Question 4

Why do we want to avoid convenience sampling?